Brooke Fitzgerald

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Department of Agricultural and Resource Economics, Colorado State University
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Education

- **Colorado State University** — *Expected 2027*
  Ph.D. Candidate in Agricultural and Resource Economics — Fort Collins, CO, USA
- **Hampshire College** — *2018*
  B.A. in Data Science and Applied Mathematics — Amherst, MA, USA

Research Areas

Environmental Economics, Air Pollution, Wildfires, Machine Learning, Remote Sensing

Working Papers

"The Price of Staying In: Estimating Wildfire Smoke Avoidance Costs Using Comprehensive US Mobility and Spending Data" with Jude Bayham, Alexandra Hill, and Jesse Burkhardt. *Submitted.* [Link]

"Measuring Wildfire Smoke and $PM_{2.5}$: Implications for Estimates of Economic Damages" with Jesse Burkhardt, Alexandra Hill, and Jude Bayham. *Preparing for Submission. Draft Upon Request*.

Selected Works in Progress

"Is that Smoke in the Air? Estimating the Global Economic Cost of Smoke Exposure Using Machine Learning" with Jesse Burkhardt, Kimberly Corwin, Yawen Guan, Emily Fischer, and Jeffrey Pierce.

"Estimating the Dual Impact of Smoke Exposure on Global Crop Yields Using Machine Learning" with Jesse Burkhardt, Kimberly Corwin, Yawen Guan, Emily Fischer, and Jeffrey Pierce.

"Firefighter Retention: Evidence from Administrative Earnings and Employment Data" with Jude Bayham, Judhajit Chakraborty, and Alisha Sharma.

Non–Peer-Reviewed Publications

"Engine Icing Data-An Analytics Approach." with Ashley Flegel. ***NASA Technical Report***. No. GRC-E-DAA-TN40097, May 2017. [Link].

Invited Talks and Conference Presentations (* = scheduled, - = pre-2023)

2026 Invited Seminar (Colorado School of Mines), CU Environmental and Resource Economics Workshop (Vail), Environmental and Resource Economics Lab Seminar (Colorado State University)*, ASSA Annual Meeting (Philadelphia)

2025 UC Berkeley Grad Camp in Energy and Environmental Economics (Berkeley), AAEA Annual Meeting (Denver), AERE Annual Meeting (Santa Ana Pueblo), The Workshop for Environmental Economics and Data Science (Portland), Front Range Energy and Environmental Economics Workshop (Boulder), Environmental and Resource Economics Lab Seminar (Colorado State University), Joint University of Wyoming and Colorado State University Graduate Research Seminar (Colorado State University)

2024 AAEA Annual Meeting (New Orleans), Western Fruit and Horticultural Society Annual Meeting (Grand Junction), Environmental and Resource Economics Lab Seminar (Colorado State University), Joint University of Wyoming and Colorado State University Graduate Research Seminar (Colorado State University)

2023- Western Agricultural Economics Association Annual Meeting (Whistler, Canada), Western Fruit and Horticultural Society Annual Meeting (Grand Junction), Environmental and Resource Economics Lab Seminar (Colorado State University), Society for Industrial and Applied Mathematics Conference on Data Mining (Calgary)

Grant and Fellowship Activity

2025 *NASA* - Future Investigator. Estimating the Dual Impact of Wildfire Smoke on Global Crop Yields Using Machine Learning and Satellite Data. $134,524 - **Awarded**

2025 Berkeley/Sloan Summer School in Energy and Environmental Economics Travel Fellowship - $750

Patents

2022 "Systems and methods for managing and controlling electronic messaging campaigns." U.S. Patent Application 16/930,916, with Kyle Tate and Pascal Potvin.

2022 "Systems and methods for controlling electronic message transmissions." U.S. Patent 11,368,422 with Mary-Juen Sohn and Tengke Xiong.

2022 "Systems and methods for dynamic messaging campaign." U.S. Patent 11,270,355 with Kevin Clark, David Cornu, and Jacinthe Ricard.

Honors and Awards

2022-2026 Graduate Teaching / Research Assistant – Department of Agricultural and Resource Economics, CSU

2023-2025 Best Graduate Poster – Department of Agricultural and Resource Economics, Colorado State University

2018 Lorna M. Petersen Prize for Collaborative Research – 5 College Consortium, Amherst, MA

2017 Best Hampshire College Statistics Student – 5 College Consortium, Amherst, MA

Professional Service

Co-Organizer of the PhD Economics Seminar at the University of Wyoming and Colorado State University – 2023-2026, Graduate Liaison for Women in Economics in the Department Agricultural and Resource Economics at CSU – 2024-2026, GSC Representative for the Department Agricultural and Resource Economics at Colorado State University – 2025

Teaching Experience

2025-2026 Colorado State University
UG: Natural Resource Economics, Module on Wildfire and Smoke (2x)

pre-2022 Five College Consortium and Hampshire College
UG: Frontiers in Biomathematics, Module on Modeling Scientific Processes

Materials Developer

2024-2025 Colorado State University
PhD: Econometric Theory II (2x)

Teaching Assistant

2022-2025 Colorado State University
PhD: Econometric Theory II (2x)
UG: Energy Economics, Agricultural Policy, Mechanical Engineering Data Analysis in R

pre-2022 Hampshire College
UG: Calculus I & II, Dynamical Systems, Introduction to Data Science, Frontiers in Biomathematics

Professional Memberships and Skills

American Economic Association, Association of Environmental and Resource Economists

- **Coding:** Python, R, Stata, SQL, MATLAB, Javascript, C++, LaTeX
- **Languages:** English (native); Portuguese, French (conversational); Spanish (intermediate)
- **Econometrics and Causal Inference:** Diff-in-Diff, Synthetic Control, Causal Forests
- **Machine learning:** Deep learning, unsupervised learning, tree-based methods
- **Big data analysis:** Parallel and cloud computing, PySpark, Arrow
- **Software engineering:** version control, web application development

References

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